UNIT 3 · 7 hours

Gain Margin and Phase Margin



Frequency Domain Analysis

Control Systems (22EEG53)
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Learning objectives

- Define gain and phase crossover frequencies
- Define gain margin and phase margin
- Determine margins from a Bode plot and analytically
- Assess stability from margins

Syllabus context

Frequency domain analysis: frequency domain specifications peak resonance, resonant frequency, bandwidth and cut-off rate; correlation between time and frequency responses for second-order systems; Bode plot, gain margin and phase margin.

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Definitions

Gain crossover ωgc

Frequency at which $|G(j\omega)H(j\omega)| = 1$ (0 dB).

Phase crossover ωpc

Frequency at which $\angle GH = -180°$.

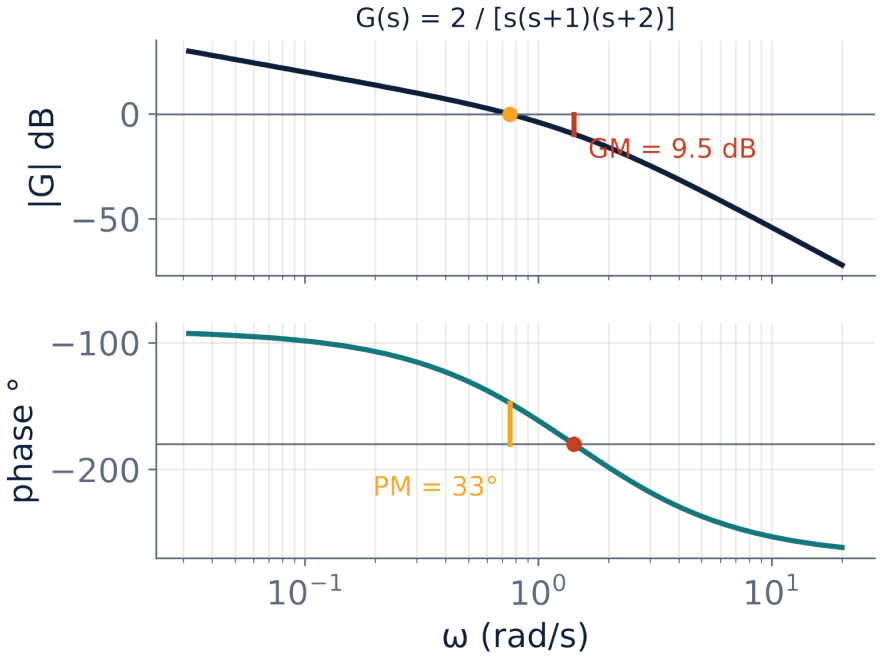
GM and PM

$PM = 180° + \angle GH(j\omega gc)$. $GM = -20 \log|GH(j\omega pc)|$ (dB), or $1/|GH(j\omega pc)|$.

Stable closed loop (minimum phase): both GM and PM positive.

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Margins on the Bode plot



- G(s) = 2/[s(s+1)(s+2)].
- ωgc = 0.75 rad/s: PM = 32.7°.
- ωpc = 1.41 rad/s: GM = 9.5 dB.
- Both positive ⇒ stable.

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Interpretation

Condition	Meaning
GM > 0 dB and PM > 0°	Stable
GM = 0 dB and PM = 0°	Marginally stable (sustained oscillation)
GM < 0 dB or PM < 0°	Unstable
Good design	GM ≥ 6 dB (≥ 2×); PM = 30°–60°
Large PM	Well damped; small overshoot

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Numerical: margins analytically

Problem $G(s)H(s) = K/[s(s+1)(s+2)]$ with $K = 2$. Find ω_{pc}, GM, ω_{gc} and PM.

- Phase $= -90° - \tan^{-1}\omega - \tan^{-1}(\omega/2) = -180° \Rightarrow \omega^2 = 2 \Rightarrow \omega_{pc} = 1.414$ rad/s
- $|G(j1.414)| = 2/(1.414 \times \sqrt{3} \times \sqrt{6}) = 2/6 = 1/3 \Rightarrow$ GM $= 20 \log 3 = 9.54$ dB
- $|G| = 1$ at $\omega_{gc} = 0.748$ rad/s; phase $= -90 - 36.8 - 20.5 = -147.3°$
- PM $= 180 - 147.3$

Answer: GM = 9.54 dB, PM = 32.7°

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Using the margins in design

- Gain K shifts the magnitude curve up/down without changing the phase: PM and GM change.
- To get the required PM: find ω where phase = −180° + PM and set $|G| = 1$ there to fix K.
- Phase margin ≈ 100 ζ links to the transient overshoot.
- Compensators (Unit 5) reshape the Bode plot to meet GM and PM.

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Key takeaways

- ω_{gc}: $|GH| = 1$; ω_{pc}: $\angle GH = -180°$
- $PM = 180° + \angle GH(\omega_{gc})$; $GM = -20 \log|GH(\omega_{pc})|$
- Stable if GM > 0 dB and PM > 0°
- Typical design: GM ≥ 6 dB, PM 30–60°

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Practice questions

Q1. Define GM and PM and state the stability conditions.

Q2. For K/[s(s+1)(s+2)] find K for marginal stability.

Q3. Find PM for G = 10/[s(s+2)].

Q4. Explain how changing K changes the margins.